

of the challenges and issues associated with LoRaWAN implementation in a white paper published recently. Areas demanding special attention in IoT solutions are security, interoperability and manageability of the solution. With such an open network, security becomes a relevant factor.

A report from HP brings to light the state of current systems. It reads, systems at present allow passwords as simple as 123456. With upgraded security in the area, we can possibly look to a future with open networks becoming the primary medium of communication. The paper also reveals dependence on out-of-the-box solutions. Twenty nine per cent of companies look to IoT Cloud platforms to lead their development projects.

"It is important to follow good practices like using secure Internet protocols like transport layer secu-

urity, sanitising your inputs, ensuring the firmware is updated to the latest version and so on," says D'Costa. Present LoRa systems use two static keys that are stored in the device and in the gateway. It could be easy to duplicate the key with a device that is not under surveillance. Also, storing keys on a device in a secure manner is quite complicated. Using a secure element hardware security module looks to be an interesting solution, with an increase in complication and expense. Advanced encryption standard-cipher based message authentication code, or AES-CMAC protocol, which includes encryption and integrity, is used for communication between the device and the gateway.

Where LoRaWAN is headed

Major IoT applications require only transmitting tiny bits of data to monitor remote devices. Mobile

systems are not geared for battery efficiency or moving little bits of data inexpensively. So a low power wide area network is required for such applications. According to Digitimes research, "LoRa Alliance in June 2015 released LoRaWAN 1.0 in competition with NB-IoT. As of end of March 2016, LoRaWAN had been commercially deployed in 13 countries and trialled in more than 60 other countries."

In a more regional update, "Shortly, we are going to deploy LoRa for some more areas in remote telemetry, though heavily interfering atmosphere is a challenge, like bringing remote cooling tower or individual cell temperature data to the control room, remote ash slurry line temperatures, etc," says Bera. He continues, "We will place the receiver at the height of the boiler, where it will be in line-of-sight for all respective senders." **EFY**



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